

Why there are more traders than manufacturers

India continues to import a significant amount of components every year. According to the released statistical figures, India is importing as much as 95 per cent of its components for the electronics industry.

Components manufacturing industry in India has not yet taken off despite the fact that there are many multinational companies which have set up manufacturing plants for electronics products in the country.

The fact that there is no duty on importing components makes component distribution more advantageous than manufacturing in the country. That explains well why there are more traders than manufacturers in India.

We see key growth in the automotive, telecommunications, mobile phones, set-top boxes and tablet PC markets. For the start, the key strategy for component distributors in India may be the promotion of automotive-grade, high-reliability and advanced technology components, which have to be imported from advanced nations with the technologies to produce such high-end products.

It will require the local electronic components manufacturers a long time to reach that level.

Apart from the fact that there are too many distributors fighting for the same market, India is also one of toughest and most cost-competitive markets in the world. Despite the volume being small, the price level expected by the local market is similar to China's.

Looking at the existing low penetration rates in many electronics applications, we think there is still room for growth in the demand of components in the country if the manufacturing industry manages to take off in the country. The growing number of middle class households in the coming years is going to translate into larger demand for electronics products. For instance, the penetration of smartphones is highly likely to increase significantly.

— **Mikio Oki**, managing director, **Murata Electronics (India)**

is largely met through imports from countries like China, Taiwan and parts of Europe.

However, "The demand for wire-wound resistors has slowly started declining because of their size, bulkiness and weight. These resistors are not available in true chip form (a chip without moulding), and therefore are not usable in applications where weight and size limitations demand precision in the chip resistor format," informs O.P. Nangia, CEO, Stead Electronics.

Unavailability of raw materials, high initial investment cost and rising infrastructure cost are some of the restraints that are negatively impacting growth in the resistor market in India. However, many manufacturers have of late realised the high revenue returns in this business.

New potential areas. Miniaturisation of electronics products, which is expected to grow significantly over the next few years, will require chip resistors. Chip resistors have low resistance and low temperature coefficient of resistance, making these suitable for use in power supply circuits.

Talking of application sector

growth, telecommunications, automotive, and IT and office automation are expected to witness a high demand for resistors in the future.

Capacitors: Ceramic type largely imported

The overall market size for capacitors in India was around \$269 million in FY 2009-10, which is estimated to have grown to \$346 million in FY 2011-12. Local manufacturing accounts for only 18 per cent of the total market. The rest of the demand is met through imports. Consumer durables (35 per cent) and telecom (20 per cent) are two of the largest demand drivers for capacitors in India.

Most Indian manufacturers largely produce film and electrolytic capacitors. Ceramic capacitors are largely imported. Lack of availability of ceramic, coupled with lack of technology to produce cost-effective capacitors, hinders the production of ceramic capacitors in India.

Film capacitors manufactured in India are also exported to countries like Europe, Hong Kong, Thailand, Philippines, Malaysia, South Korea and Middle East. The export market for

capacitors was estimated at \$40 million in FY 2009-10.

Capacitor manufacturers are concerned about the increasing cost of raw materials and the decreasing revenue realisation on the final component due to the availability of cheaper import options.

New potential areas. Currently, several experiments are underway to design and develop capacitors that can hold charge for a long period. This will enable them to be used in several other fields.

With the world moving towards developing energy-efficient devices, super capacitors and ultra capacitors are gaining importance globally. Super capacitors, with a lower discharge loss, are expected to replace traditional storage batteries. Ultra capacitors are expected to help in providing fuel-efficient transport as well as renewable energy.

Niobium-oxide (NbO) capacitors are being increasingly preferred over conventional aluminium electrolytic capacitors as these capacitors have stable electrical parameters like capacitance, ESR and leakage current. These features make NbO capacitors more suitable for use in audio devices, PCs, automobiles and miniature devices like portable digital sound processors, echoes, gates, microphone amplifiers and DVD players.

Wound components: Growing steadily

The Indian wound component market has been witnessing a steady growth for the last few years and is estimated to have reached a value of \$2035 million in FY2011-12, largely driven by growth in consumer durables, medical electronics and industrial electronics. Indigenous manufacturing contributes close to 30 per cent of the demand.

It is estimated that there are more than 200 wound component manufacturers in India, most of them being SSI units. Most of the inductor demand is met by indigenous manufacturing, while chokes and transformers are